

Katalin Berlow

Curriculum Vitae

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Research Interests

My research focuses on descriptive combinatorics and measured group theory with applications to ergodic theory and geometric group theory. I work with the broader goal of understanding how combinatorial structure encodes dynamical and group-theoretic information.

Appointments

Stephen E. Warschawski Visiting Assistant Professor 2026–2029
University of California, San Diego, La Jolla, CA

Education

Ph.D. in Mathematics, 2021–2026
University of California, Berkeley, Berkeley, CA
Advisors: Andrew Marks and Robin Tucker-Drob

B.S. in Mathematics with Honors 2017–2021
Carnegie Mellon University, Pittsburgh, PA

Honors and Fellowships

Simons Dissertation Fellowship in Mathematics 2025
NSF Graduate Research Fellowship 2021
Fulbright Fellowship, *declined* 2021
Benter Foundation Mathematics Scholarship 2021
Duluth REU 2020

Research Publications

In Preparation

- [1] **A Measurable Analog to Euler’s Formula in pmp Graphs** (in preparation).
- [2] **Equivalence Relations Acting on Bundles of Trees** (in preparation).
with R. Chen, A. Tserunyan, and R. Tucker-Drob
- [3] **Separating Complexity Classes of LCL Problems on Grids, Part 2** (in preparation).
with A. Bernshteyn, C. Lyons, and F. Weilacher

Submitted

- [4] **The Polychromatic Number of Grids** (submitted, 2025).
<https://arxiv.org/abs/2508.18559>. *with E. Hou*
- [5] **Separating Complexity Classes of LCL Problems on Grids** (submitted, 2025).
<https://arxiv.org/abs/2501.17445>. *with A. Bernshteyn, C. Lyons, and F. Weilacher*

Published

- [6] **Intersection Bodies of Polytopes**, *Beiträge zur Algebra und Geometrie* 63 (2022), 419–439.

<https://arxiv.org/abs/2110.05996>. with M. Brandenburg, C. Meroni, and I. Shankar

[7] **Analysis on Almost Abelian Lie Groups**, *Journal of Mathematical Sciences* (2022).
<https://arxiv.org/abs/2004.04369>. with M. Almora-Rios et al.

[8] **Restricted Stacks as Functions**, *Discrete Mathematics* 334 (2021).
<https://arxiv.org/abs/2008.01164>

Selected Talks

Invited Talks and Workshops

TBD	Jan 2027
Joint Mathematics Meetings	
Workshop on group actions and equivalence relations	July 2026
Rényi Institute, Budapest, Hungary	
Local Problems on Grids: FIID/FFIID Separations and a Failure of Baire Shift Universality	Apr. 15, 2026
Caltech Logic Seminar	
Measurable analog of Euler's formula for pmp graphs	Mar. 2026
UT Austin Groups and Dynamics Seminar, Austin, TX	
Separating Complexity Classes of LCLs on Grids	Mar. 2026
AMS Spring Western Sectional Meeting, Boise, ID	
A Measurable Analog to Euler's Formula	Feb. 2026
Rényi Borel Combinatorics Seminar, Budapest, Hungary	
Separating Complexity Classes of LCLs on Grids	Oct. 2025
North American Descriptive Set Theory Conference, University of Maryland	
Separating Complexity Classes of LCLs on Grids	May 2025
CanaDAM, Ottawa, Canada	
Separating Complexity Classes of LCLs on Grids , South Eastern Logic Symposium, University of Florida	Mar. 2025

Teaching

Main Instructor

MATH 109: Mathematical Reasoning, UC San Diego	Fall 2026
MATH 110: Abstract Linear Algebra, UC Berkeley	Summer 2025

Teaching Assistant / Graduate Student Instructor

MATH 135: Set Theory, UC Berkeley	Spring 2026
MATH 142: Elementary Algebraic Geometry, UC Berkeley	Spring 2026
MATH 113: Abstract Algebra, UC Berkeley	Fall 2025
MATH 1B: Calculus II, UC Berkeley	Spring 2023
MATH 54: Linear Algebra and Differential Equations, UC Berkeley	Fall 2022
MUSA 74: Transition to Upper-Division Mathematics, UC Berkeley	Fall 2021
21-127: Concepts of Mathematics, Carnegie Mellon University	Spring 2020
33-142: Physics II for Engineers, Carnegie Mellon University	Summer 2018

Organizing and Outreach

Mentorship

Visiting Research Mentor, Duluth REU in Combinatorics 2021–2024

Mentor, UC Berkeley Directed Reading Program 2021–2026

Topics: extremal and probabilistic combinatorics, descriptive set theory, equidecompositions, amenable groups

Seminars Organized

Distributed Algorithms and Borel Combinatorics Seminar, UC Berkeley Fall 2023

Descriptive Set Theory Seminar, UC Berkeley Spring 2022

Topics in Mathematics, virtual seminar 2021

Professional Service

Reviewer, *Discrete Mathematics*

Expository Writing and Miscellaneous

Expository Writing

Amenable groups acting on trees, expository note on website

Riddles, expository collection on website

Miscellaneous

Erdős number: 3

Languages: English (native); Hungarian (conversational)